What changes an asteroid’s path?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how gravity can change velocity without contact.

SUCCESS CRITERIA - I can...

1. Draw velocity tangent to a path and force toward the attracting body.
2. Distinguish motion from a change in motion.
3. State what a prediction of straight-line motion leaves out.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gravity” and “acceleration” to explain your answer. Sentence starters are optional.

gravity: an attractive interaction between masses

acceleration: rate of change of velocity

model: a drawing, explanation or equation for a physical situation

01 Warm-up: bring back an earlier idea

A **3 kg** cart starts from rest under **15 N** net force right for **2 s**.
1. **Find acceleration** and final velocity.
2. **State one thing you** are treating as true.

Word help: net force = the vector sum of all forces on one object

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

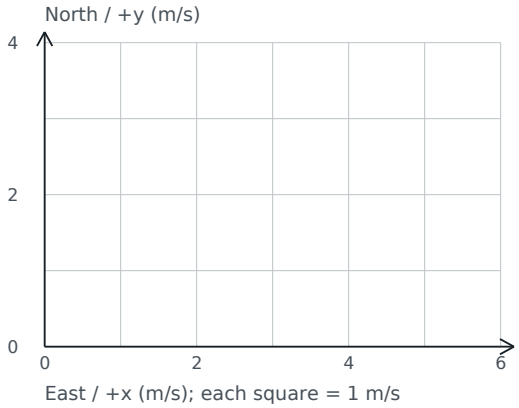
02 Worked example: explain the prediction

Reference: $a = F_{net}/m$.
Gravity attracts masses and can change speed, direction or both. Our asteroid scenario is fictional.

Worked reasoning: A fictional asteroid has a velocity arrow pointing right. A brief upward force gives it an upward change in velocity. The new velocity combines the original rightward motion with this change.

Your turn: A spacecraft is initially moving right. A short thruster burn adds an upward change in velocity.
1. **Draw the original velocity** and the change head-to-tail.
2. **Draw the final velocity** from the starting tail to the ending tip.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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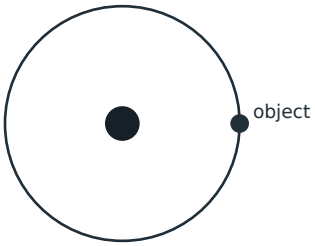
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

1. **Sketch the velocity** and gravitational-force arrows at the marked point on the orbit.
2. Why do they point in different directions?

Sentence starter: Both … . They differ in … because … .



The center is the attracting body; the small dot on the circle is the orbiting object.

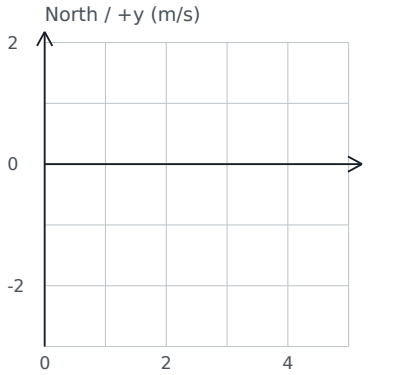
Marzano check - circle: 1 2 3

04 Apply the physics

In a flight simulator, a probe moves right at **4 m/s**. After a **1 s** thruster burn, it also moves down at **1 m/s**.

1. **Add the components** and final velocity to the grid.
2. **Explain why the probe still** moves right.

Sentence starter: I think … . I can point to …, which shows …, so … .



East / +x (m/s); each square = 1 m/s

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says space has no gravity because astronauts float. **Explain how orbiting astronauts can** be falling.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

With a fixed net force, a **5 kg** system accelerates at **14 m/s²**. **Predict acceleration** for a **10 kg** system. **State a control needed** for this comparison.
Use the table to keep the controlled and changed quantities distinct.

Word help: system = the object or objects chosen for analysis

Sentence starter: I predict … . I used … to make this prediction because … .

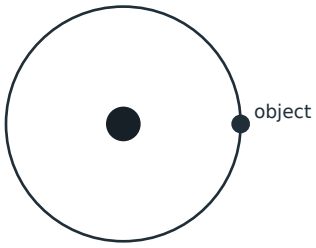
Trial	Mass	Net force	Acceleration
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

07 Exit check

Name one observation that shows velocity is changing even if speed stays the same.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A probe repeats the same thruster test, first with a short burn and then with a longer burn. Mass and thrust stay fixed.
1. **Sketch a velocity-change arrow** for each trial.
2. Which burn causes a larger change, and why?

Sentence starter: Both … . They differ in … because … .

Compare the two burns

Right is positive.



Velocity:

Acceleration / force:

Marzano check - circle: ① ② ③